7 Interdependence of Living Beings: Plants and Animals

All living things depend on other living as well as non-living things for their survival. We will learn about the interdependence of plants and animals in this chapter.

Learn about

- Interdependence of plants and animals
- Living components of the environment
- Categorization of living components
- Food chain
- Balance in nature
- Causes of imbalance in nature

Interdependence of Plants and Animals

Plants and animals are dependent on each other for survival. Plants provide animals with food, shelter, and protection. Similarly, animals help plants in dispersal of seeds, and also provide the nutrients needed by them for making food.

Dependence of Plants on Animals

Plants depend on animals for several basic needs. Animals breathe in oxygen and breathe out carbon dioxide. This carbon dioxide is used by plants to prepare food by the process of photosynthesis. When animals die, they are converted back into nutrients (by small organisms called microorganisms) and become a part of the soil. These nutrients are used by plants for growth and survival. Animals also help in pollination and dispersal of seeds.

Dependence of Animals on Plants

Animals cannot make their own food. They depend directly or indirectly on plants for food. Animals also need oxygen to breathe. They get this oxygen from plants. Plants release oxygen during photosynthesis. Plants also provide shelter and protection to animals and birds who live on trees.

Living Components of the Environment

Our environment is made up of several non-living and living components. The non-living components of the environment are soil, air, light, water, and warmth.

All living things such as plants, animals, and microorganisms (such as bacteria and fungi) form the living components. All these components depend on each other for existence. Let us learn more about the living components of the environment.

Plants

Plants produce their own food by the process of photosynthesis using carbon dioxide from the air and water from the soil. They also need sunlight and chlorophyll for photosynthesis to take place. All living organisms depend directly or indirectly on plants for food. Plants usually stay fixed in their positions and depend on the environment for their survival.

Think and Discuss

21ST CENTURY SKILLS

Think about the food you eat. Can you trace it back to a plant? How does this show the connection between plants and animals?

Plants depend on the non-living components of the environment in the following ways: They get water and nutrients from the soil. They get carbon dioxide from the air and release oxygen into the air. When plants die, they are converted back into nutrients by microorganisms. These nutrients then get mixed with the soil.

Animals

Animals cannot prepare their own food. They move from one place to another to find food. They either feed directly on plants, or on other animals.

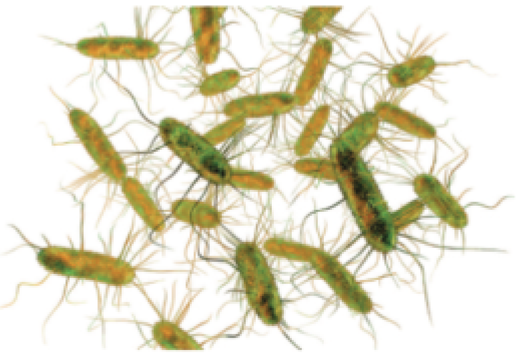
Fact File

Primary producers are also called autotrophs as they can make their own food.

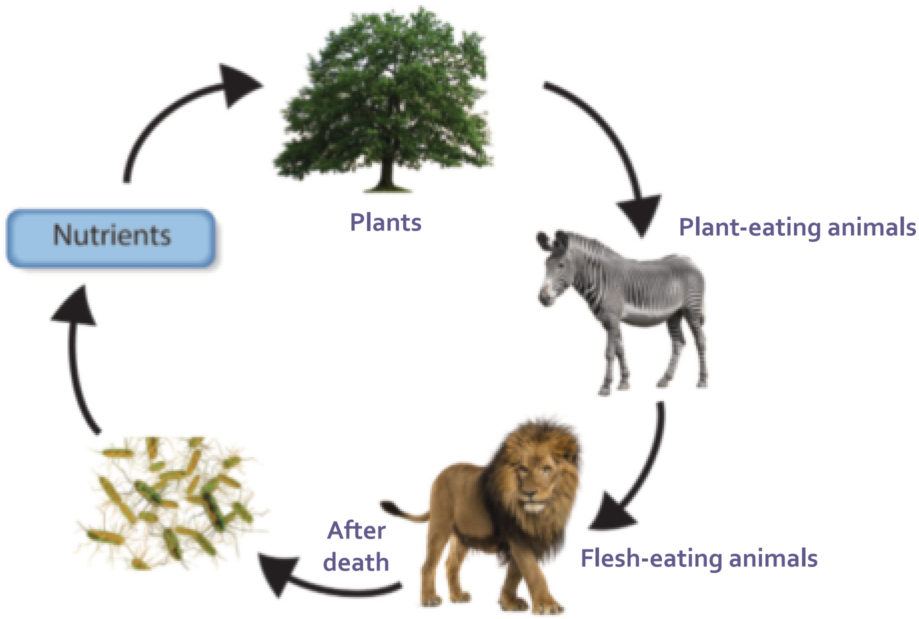
Animals depend on the non-living components of the environment in the following ways: They need water to survive. They breathe in oxygen present in air, and breathe out carbon dioxide.

Microorganisms

Microorganisms are very small organisms present in the environment. They cannot be seen with naked eyes. Most microorganisms depend on dead plants and animals to obtain food for survival. They feed on dead organisms and release the nutrients from them into the soil. These nutrients are then used by plants for growth.



Microorganisms



Interdependence in the environment

Categorization of Living Components

Depending on their roles in the environment, all living components can be put into the following categories.

Producers

All living organisms that can produce their own food are called ***producers****.* All green plants are producers because they can prepare food by the process of photosynthesis. In photosynthesis, plants take carbon dioxide from the air and water from the soil, and convert them into food in the presence of sunlight and a green pigment called **chlorophyll**. This food is used by plants for their growth and survival. Plants are then eaten by animals. Therefore, producers are the main source of food and energy to other living organisms.


Producer

Consumers

All living organisms that cannot prepare their own food, and depend on plants or other living organisms for food are called ***consumers****.* All animals are consumers. Consumers can be of the following types:

- *Animals that feed only on plants are called* ***herbivores****.* Cow, goat, sheep, elephant, and giraffe are examples of herbivores.

- *Animals that feed on other animals are called* ***carnivores****.* Some carnivores feed on herbivores, whereas some feed on other carnivores. Lion, tiger, crocodile, fox, hawk, and snake are some examples of carnivores.
- *Animals that feed on both plants and animals are called* ***omnivores****.* Bears and human beings are examples of omnivores.


Rabbit


Snake


Eagle

Examples of consumers

Scavengers

Animals that feed on dead plants and animals are called ***scavengers****.* Scavengers play an important role in keeping the environment clean by consuming dead matter. Examples of scavengers include vultures and termites. Vultures feed on dead animals, whereas termites feed on dead plants. Other examples of scavengers include earthworms, bears, and snails.


Vulture

Example of a scavenger

Decomposers

Decomposers are microorganisms such as bacteria and fungi that feed on dead plant and animal matter. Scavengers feed on the dead plants and animals and break them down into small pieces. These small pieces are then eaten by the decomposers. They break down the nutrients present in the decaying matter into simpler forms, which then get mixed with the soil. These simpler nutrients are then absorbed from the soil by plants. *The process of breaking down complex nutrients into simpler substances by decomposers is called* ***decomposition****.* Therefore, decomposers play a very important role in the cycle of nutrients, in the environment.

Questions

Fill in the blanks.

1. Plants and animals are dependent on each other for ....
2. When animals die they are converted back into nutrients by ....
3. Plants get water and nutrients from the ....
4. All living organisms that cannot prepare their own food are called ....
5. Decomposition is the process of breaking down .... nutrients into .... substances by decomposers.

Food Chain

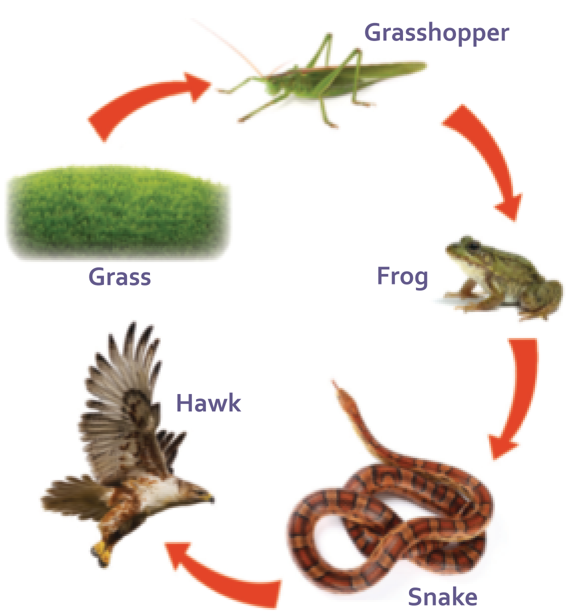
Animals cannot make their own food. They eat plants or the flesh of other animals. For example, grass is eaten by a rabbit and the rabbit is then eaten by a fox.

Fact File

Al-Jahiz was a 9th century scientist who introduced the concept of food chain.

This event forms a small chain. Let us take another example.

Grass is eaten by a grasshopper. The grasshopper is, then eaten by a frog. The frog is, again eaten by a snake and the snake is, finally eaten by a hawk. This forms a chain that links plants and different animals for food. *A chain that shows how plants are eaten by animals who are then eaten by other animals is called a* ***food chain****.* A food chain always begins with plants.



Food chain

In other words, a food chain always begins with producers (green plants). The producers are then eaten by herbivores. Herbivores are, in turn, eaten by carnivores and omnivores. Smaller carnivores may further be eaten by larger carnivores. Many food chains may interconnect, forming a **food web**. This happens when an organism can be eaten by many predators from different food chains, or when it eats many other organisms of different food chains.

Balance in Nature

> **Fact File**
>
> Food chains not only exist on lands, but under water bodies too! These chains start with very minute organisms called **planktons**.

All living organisms depend on each other for food. This interaction for food causes food chains in the environment. The addition of new animals or absence of some animals from the food chain will upset the balance in nature and may even cause other animals and plants to die. For example, if the population of carnivores in an area decreases, it will cause the population of herbivores to increase as they will no longer be hunted for food. The increased number of herbivores would eat all the plants in that area. After that, the herbivores would also die because of the absence of food. Hence, all animals and plants are essential to maintain the balance in nature.

Causes of Imbalance in Nature

Primarily, imbalance in nature can be caused due to natural causes and man-made causes. Let us learn about them.

Natural Causes

Floods Due to heavy rainfall, a large amount of water sometimes accumulates in an area. This condition is called **flood**. This may kill a large number of plants and animals present there, thus causing an imbalance in nature.

Drought During summers, when there is no rainfall for a long time, water bodies like rivers and lakes dry up. This condition is called **drought**. Large number of plants and animals die due to shortage of water.

Forest fires Sometimes lightning can cause a fire in the forest that wipes out animals and plants in large numbers. This also causes an imbalance in nature.

Land clearing

Human-Made Causes

Land clearing It is the practice of cutting down trees and removing vegetation from a forest area for farming and other human activities. Land clearing causes the death of several kinds of plants and animals. It also leads to a decrease in rainfall and an increase in the amount of carbon dioxide in the atmosphere.

Hunting It is the practice of killing or capturing animals for food, sport, or for products such as horns, leather, ivory, and fur. Excessive and illegal hunting may result in the death of various animals. For example, the number of rhinoceros killed for their horns is so high that very few of them are left in the world. There are many such animals that are left in very small numbers and may die out very soon.

Pollution It is mostly caused by human activities such as throwing wastes in large amounts on lands and in water, and release of excessive smoke and poisonous gases from factories and industries. Pollution causes land and water to become poisonous for living organisms. This may cause several animals and plants to die.


Dumping of solid wastes on land contributing to pollution

Forest fire It is the practice of burning down parts of forests to clear the land for farming or for making human settlements. Forest fires may also be caused by lightning strikes and other natural causes as already discussed before. Uncontrolled forest fires can kill all living organisms present in the forest.

Restricting Imbalance in Nature

The imbalance that is caused due to human-made causes can be minimized if we stop the exploitation of nature. The various steps for it are as follows:

- We should restrict the practice of land clearing which involves cutting down trees on a large scale. We should plant more trees to check drought and soil erosion.
- Excessive and illegal hunting of wild animals should be stopped.
- Human activities that lead to pollution of air, water, and soil should be checked.
- Burning down of forest by humans to clear land should also be stopped.
- Do not waste paper. Reuse paper whenever possible as lots of trees are cut to make paper.

Case Study

Smita, a passionate science teacher, was discussing the topic of "The Effect of Imbalance in Nature" with her students. Kriti, who loves reading magazines and newspapers, pointed out that overfishing and pollution are causing many marine animals like fish and turtles to die out. They also discussed how plastic pollution harms marine life when industrial runoff poisons the water. This pollution can impact humans who eat seafood and harm local fishing communities.

Based on the above information, answer the following questions:

a. How does plastic pollution harm marine life?

b. How can pollution in the oceans impact humans?

Wrap Up

- Plants provide animals with food, shelter, and protection. Animals aid in seed dispersal and provide nutrients for plants.
- The non-living components of the environment are soil, air, light, water, and warmth. All living things such as plants, animals, and microorganisms (such as bacteria and fungi) form the living components.
- Microorganisms are very small organisms present in the environment. They cannot be seen with the naked eyes.
- All living organisms that can produce their own food are called producers.
- All living organisms that cannot prepare their own food and depend on plants or other living organisms for food are called consumers.
- Herbivores feed on plants, carnivores feed on other animals, omnivores feed on both plants and animals, and scavengers feed on the dead plants and animals.
- The process of breaking down complex nutrients into simpler substances by decomposers is called decomposition. Decomposers like bacteria and fungi feed on dead plant and animal matter.
- Several human activities, such as land clearing, hunting, pollution, and forest fires, are harmful for the environment.

Exercises

SECTION I

REMEMBERING UNDERSTANDING APPLYING

A **Choose the correct option.**

1. When animals die, they are converted back to ....

   a. oxygen b. carbon dioxide c. sunlight d. nutrients

2. Which of these do plants depend on animals for?

   a. Nutrients b. Dispersal of seeds

   c. Carbon dioxide d. All of these

3. Which of these is a non-living component of the environment?
   a. Air b. Light c. Soil d. All of these
4. Which of these is an example of a herbivore?
   a. Goat b. Lion c. Cat d. Tiger
5. Which of these is an example of a scavenger?
   a. Elephant b. Giraffe c. Vulture d. None of these

B Assertion and Reasoning questions

1. **Assertion (A):** Plants and animals depend on each other for survival.
   Reason (R): Plants provide animals with food and shelter, while animals help plants disperse seeds.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True
2. **Assertion (A):** Scavengers compete with decomposers for food.
   Reason (R): Both feed on dead plant and animal matter.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

C Write *T* for True and *F* for False. Correct the False statements.

1. Plants do not depends on animals.
2. Plants give shelter and protection to some animals.
3. Plants, animals, and microorganisms form the living components.
4. All animals are consumers.
5. Animals that feed on both plants and animals are called carnivores.

SECTION II

REMEMBERING UNDERSTANDING APPLYING

D Short answer questions

1. How are plants and animals dependent on each other?
2. You're hiking in the woods and discover a dead squirrel under a tree. What would happen to the squirrel's body over time?
3. Based on their role in the environment, in how many categories are all living organisms divided into?
4. What are producers? Why are they important?
5. Define decomposition.

E Long answer questions

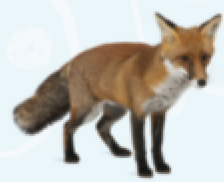
1. What do you understand by interdependence of plants and animals?
2. What are consumers? How many types of consumers are there?
3. Describe food chain with suitable examples.
4. How is imbalance in nature caused?

Picture Study

1 Complete the given picture by putting arrows in it correctly.

(i)

(ii)



(iii)

....

2 How are these living organisms categorized?

....

3 What does the given picture depict?

....

4 Write an important feature of this chain.

....

My Learning Corner

A Think about

Bunny is afraid of lizards in his home. He suggests to his mother that they are a nuisance for us. Therefore, it will be better if they all die. What is your opinion? [Hint: What does a lizard feed upon? What will be the effect on them if there are no lizards?]

....

....

B Try out

1. Organize a group activity: Divide the whole class into groups of 4 or 5 students. Each group will visit the school's herbal garden, nearby garden, or on a field trip along with their teacher.

   Write the names of living organisms you see in the garden and categorize them in two ways:

   i. Producers, herbivores, carnivores, and omnivores

   ii. Producers, consumers, and decomposers

   Write this activity in your Science scrapbook.

2. Make a model of a food chain using cardboard, chart paper, black sketch pen, gum, and using coloured pictures of the living organisms.
3. Conduct a class discussion on what would happen if one of the producers or the consumers in the food chain disappeared.
4. Now, you are aware of how forest fires occur. On the political map of India, mark the states where forest fires are quite frequent.

STEM

SCIENTIFIC SKILLS

ART INTEGRATION

SUBJECT INTEGRATION

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• differentiate between plants (producers) and animals (consumers) based on their features.	
• infer why plants can make their own food.	
• cite examples of producers and consumers.	
• classify living beings as producers and consumers.	
• explain the food chain and food web with examples.	
• identify decomposers and scavengers with examples.	
• explain the causes of imbalance in nature.	
• analyze the effect of hunting and forest fires on the environment.	

Worksheet

Observe the picture based on 'Interdependence of living organisms' given below with empty labels. Complete these labels with the appropriate word or phrase given in the word box below.

CARNIVORES	NUTRIENTS released in soil	HERBIVORES
After death	PRODUCERS	MICROORGANISMS
Taken up by